

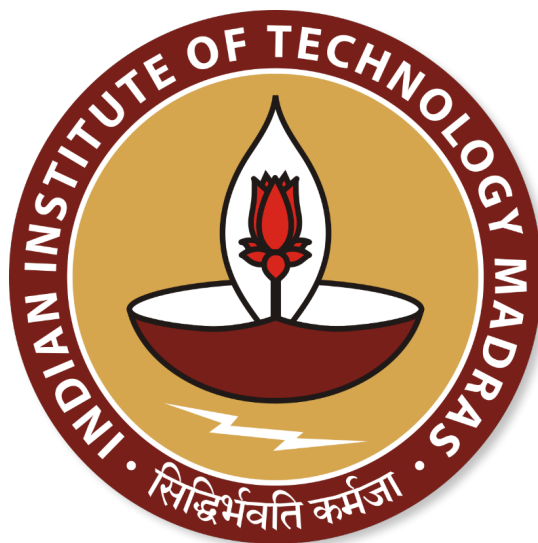
Supply Chain and Inventory Strategies to Drive Profitability in a Juice Shop

MID TERM REPORT

Submitted by

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Contents

S.no	Theme for the page	Page
1	Executive Summary and Title	1
2	Proof of originality	1
3	Metadata	2
4	Descriptive Statistics	3-4
5	Analysis Methods	5-6
6	Results and finding	7-8

1.Executive Summary

I am working with a small shop named “Food&Drink” located at Plahi Gate, Phagwara, Punjab. It is B2C business and deals in the segment of selling all types of juices, shakes, snacks, coffee. Shop was opened in May 06, 2024 and I have collected 6 months of sales and investment data from mid July onwards manually in a single spreadsheet with the support of owner.

In this document I have taken basic descriptive statistics explaining the picture of sales varies according to weather. Sales goes from 58k in Aug to 42k in Sep and around 28k in Oct, Nov and 26k in Dec and Jan. There is very big difference between sale of food and drinks. I have given recommendations to owner about increasing more varieties in Food to get maximum sales in food compare to drink because there is very big difference that caught in the sales trends of food and drinks. Food items is very less sold during summers. I have also compare statistics of all SKUs and draw their sales graphs to get most sold item and focus on it to improve profit and introduce its varieties in future.

2.Proof of originality

-  [Click here for Interaction video with owner](#)
-  [Click here for Picture of shop with owner](#)
-  [Click here for Shop Data.](#)
-  [Click here for Letter head from owner](#)



3. Metadata

A. **Master Data:** It contains columns named as “SKUs” ,”Price” and has 6 rows for SKU price.

ALL SKU Prices				Price 2024
Milk Soda	Lemon Soda	Corn M		30
Oreo Shake	Black Current Shake	Apple Juice	Hot Coffee L	50
Veg Sandwich	Soya kabab Sandwich	Blue Berry Shake	Chocolate Shake	60

B. **Sales Data:** There are around 30 columns in actual sales data for each month from August to January and there are 6 sub sheets for sales for each month. Each cell in row of Time Intervals contain total units of sales of each SKU.Each cell in front of Date contain total units sold as the sum of all intervals on each day. Total sales column has units sold for each row per interval and per day.

Date & Timing	Snacks	Sandwich	Juice	Coffee	Shakes	Soda Drink	Total
01-MM-YY	25	8	23	9	11	33	109
10AM-1PM	4	1	4	2	3	10	24
1PM-4PM	7	2	5	4	2	8	28
4PM-7PM	5	2	6	2	2	9	26
7PM-10PM	9	3	8	1	4	6	31
02-MM-YY	...	...	...	...	...	...	...

C. **Investment Data:** There are 6 sub sheets of investment data from August to January.Each cell in investment sheet contain total cost on particular items used for making juice, shakes, sandwiches.

Date	Banana	Milk	Soda	Patty	Disposable	New Item	Total cost
01-MM-YY	100	600	200	100	0	0	1000
02-MM-YY	100	600	200	150	200	0	1250
Total	200	1200	400	250	200	0	2250

D. **Survey Data:** It is provided in single sheet that is connected to google form and filled automatically as someone fills the form using QR code pasted on customer table.It has 9 fields that can vary along the time.While filling the form we have collected their email for future reasons to connect with themThere are 5 questions asking which food or drink they have it today in the shop and then followed by Rating for todays order. Rating Field contain stars from 1 star to 5 star (bad to excellent). Feedback field is optional field for those who want to write something else in short to upgrade in the shop menu.

Timestamp	Email	Q1	Q2	Q3	Q4	Q5	Rating	Feedback
9/25/2024 23:45:52	abc@gmail.com	Yes	No	No	Yes	No	4 star	Good
9/28/2024 20:25:04	mn@gmail.com	Yes	No	No	No	Yes	5 star	Excellent

4.Descriptive Statistics

Statistical	Baked	Grill Sandwich			Sweet Corns		
Parameters	Patties	Paneer Corn	Special Veg	Soya kabab	Small	Med	Large
Mean	160.64	96.95	86.44	20.87	15.07	27.39	2.32
Median	160.00	80.00	60.00	0.00	0.00	0.00	0.00
Mode	200.00	0.00	0.00	0.00	0.00	0.00	0.00
S. Variance	10479.08	11252.15	9202.61	1835.57	616.54	1846.04	88.66
Kurtosis	2.61	1.87	1.92	8.18	8.39	5.58	13.34
Skewness	0.96	1.28	1.34	2.64	2.43	2.17	3.87
Minimum	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Maximum	580.00	480.00	480.00	180.00	140.00	210.00	40.00
Sum	25220.00	11440.00	10200.00	480.00	1040.00	1890.00	160.00
Count	157.00	118.00	118.00	23.00	69.00	69.00	69.00

Food (MOST SOLD SNACKS)

Paneer sandwiches goes from 16 units to 37 units from Sep to Oct

V eg sandwiches goes from 8 units to 28 units from Sep to Oct

Patty comes down from 333 units to 187 units to 167 units from Aug to Sep to Oct.

Statistical	Soda Drinks			Juices				
Parameters	Lemon Soda	Milk Soda	Lemonade	Musambi Juice	Pineapple Juice	Carrot	Apple	Mix Juice
Mean	91.56	287.10	30.62	78.59	17.78	38.64	8.41	189.50
Median	60.00	270.00	20.00	50.00	0.00	40.00	0.00	160.00
Mode	0.00	0.00	0.00	0.00	0.00	0.00	0.00	120.00
S. Variance	8628.06	59758.51	1610.86	8084.79	1187.42	2591.23	683.30	21262.64
Kurtosis	1.95	0.54	2.69	1.47	2.72	4.20	16.25	-0.17
Skewness	1.33	0.89	1.65	1.31	1.93	1.83	3.83	0.73
Minimum	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Maximum	420.00	1110.00	180.00	400.00	120.00	240.00	150.00	600.00
Sum	8790.00	30720.00	2480.00	7230.00	960.00	2280.00	900.00	30320.00
Count	96.00	107.00	81.00	92.00	54.00	59.00	107.00	160.00

Statistical	Shakes									
Parameters	Mango Shake	Banana Shake	Black Current Shake	Blue Berry Shake	Butterscotch Shake	Custom Shake	Chocolate Shake	Oreo	Papaya Shake	Strawberry Shake
Mean	94	145	15	11	20	15	77	61	30	46
Median	60	80	0	0	0	0	0	50	0	40
Mode	0	0	0	0	0	0	0	0	0	0
S.Variance	10337	23970	1387	1038	1387	822	12461	3709	2621	3874
Kurtosis	-1	0	15	13	3	2	5	0	7	6
Skewness	1	1	3	3	2	2	2	1	2	2
Minimum	0	0	0	0	0	0	0	0	0	0
Maximum	320	600	250	180	160	120	660	250	320	360
Sum	2640	23120	1650	1460	1280	2040	12300	9800	4760	7280
Count	28	160	107	134	63	132	160	160	160	160

Statistical	Cold Coffee				Hot Coffee		
Parameters	Normal	Chocolate	Custom	Butterscotch	Small	Med	Large
Mean	22.75	32.75	7.14	44.50	10.94	206.09	10.14
Median	0.00	0.00	0.00	0.00	0.00	210.00	0.00
Mode	0.00	0.00	0.00	0.00	0.00	270.00	0.00
S. Variance	2347.11	2110.63	598.94	6359.50	2070.25	14847.70	630.86
Kurtosis	7.28	1.12	17.87	7.20	15.36	-0.37	5.60
Skewness	2.62	1.35	4.08	2.45	4.06	0.31	2.51
Minimum	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Maximum	240.00	200.00	120.00	480.00	220.00	510.00	100.00
Sum	3640.00	5240.00	200.00	7120.00	580.00	14220.00	700.00
Count	160.00	160.00	28.00	160.00	53.00	69.00	69.00

Drinks (MOST SOLD DRINKS)

Milk soda also decreases as weather changes from hot to cold, it goes from 532 to 305 to 194 units of sale from Aug to Sep to Oct respectively.

Lemon soda also decreases as weather changes from hot to cold, it goes from 137 to 76 to 83 units of sale from Aug to Sep to Oct respectively.

5. Analysis Methods

Analysis method can be broken into 4 different steps:

- A. Data Collection
- B. Data Cleaning
- C. Descriptive Analysis (What happened?)
- D. Data Visualisation
- E. Diagnostic Analysis (Why did it happen?)
- F. Prescriptive Analysis (What should we do?)

A. Data Collection

I have used google sheets with editing access to both the owner and me. We daily noted both sales and investment data together of 6 months from 1-Aug-2024 to 31-Jan-2024. Data has been manually feed into the spreadsheet without usage of any app or ERP system. I have collected sales data as the number of units sold per SKU, but on the other hand Shop owner has daily noted investment cost per day when he buy raw material from the market. After this I have created one master sheet that contains price of each SKU and I have group the SKUs with same price and write the price of group of SKUs in the next field.

I have also taken survey data from customers using QR code and redirected it to google form that linked with another sheet. I have generate this QR code from Canva that linked to the google form created by me.

B. Data Cleaning

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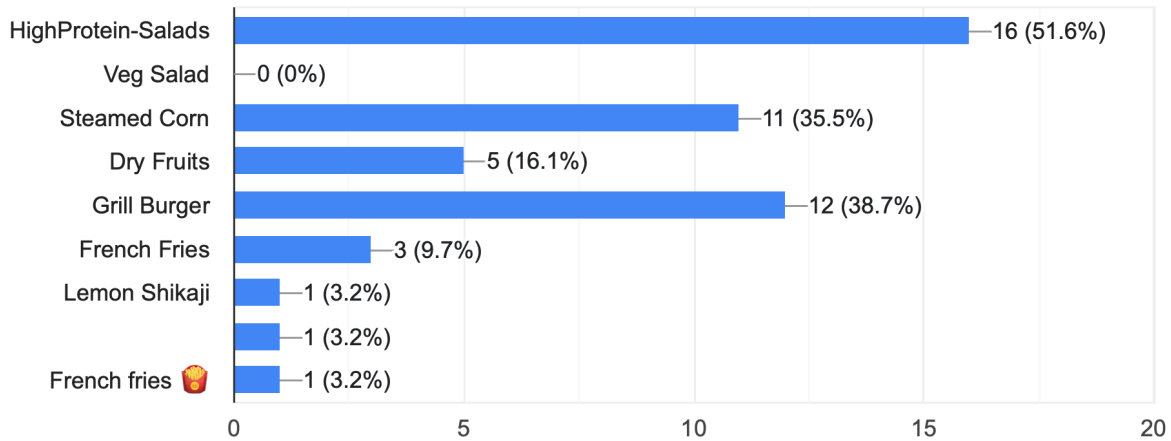
C. Descriptive Analysis

This is the very important step in data analysis because it gives us picture of data

6.Results and finding

Q3: Any thing you expect us to add in the menu in food items for winters ? (Multiple Options can be selected)

31 responses



Any thing you expect us to add in the menu in drink items for winters?

31 responses

